

Bioinformatics



Sequence Alignment

- Sequence alignment is used in bioinformatics and molecular biology to compare biological sequences(DNA, RNA, or proteins).
- Understand their similarities and differences.
- These similarities often reveal **functional, structural, or evolutionary relationships** among genes or proteins.

Main Reasons Why Sequence Alignment Is Used

- To find evolutionary relationships.
- To identify similar or conserved regions
- To predict the function of unknown sequences
- To compare protein structures and functions
- To assist in drug design and vaccine development

In Simple Terms

Sequence alignment helps scientists **understand relationships, predict functions, and discover important biological patterns** by comparing genetic or protein sequences.

Types of Sequence Alignment

1. Pairwise Alignment

Compares **two sequences** at a time.

Used to find the best possible alignment between them.

Two main methods:

Global alignment (Needleman–Wunsch algorithm):

- Aligns entire sequences from end to end.
- Suitable when sequences are of similar length and overall similar.

Local alignment (Smith–Waterman algorithm):

- Finds regions of local similarity.
- Useful when sequences are partially similar or contain conserved regions.

Types of Sequence Alignment

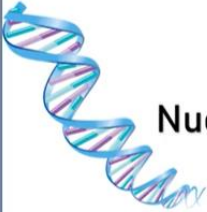
2. Multiple Sequence Alignment (MSA)

- Aligns three or more sequences simultaneously.
- Used to identify conserved motifs, phylogenetic relationships, and functional domains.

Common tools: **Clustal Omega**, **MUSCLE**, **MAFFT**, **T-Coffee**.

Sequences & Alignment

Sequence similarity and identity

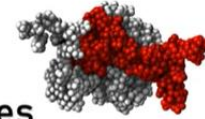


DNA

Nucleotide Base Sequences

Protein

Amino acid Sequences



Why align the sequences ?

Functional Relationship



Structural Relationship



Evolutionary Relationship



Volume

Why align sequences?

□ Functional Relationship

Question: Do two genes/proteins perform similar functions?

Example:

Protein A: M K L A V G

Protein B: M K L A I G

Most amino acids are the same.

Therefore:

- Their structures may be similar.
- Their functions may be similar.

Example:

Human insulin and animal insulin have similar sequences, so they perform similar roles.

Why align sequences?

❑ Structural Relationship

Question: Do two proteins have similar 3D structures?

Sequence determines structure.

If two proteins have similar sequences,
They may fold into similar shapes.

❑ Evolutionary Relationship

Question: Did two organisms come from a common ancestor?

Example:

Human DNA: A C G T A C

Chimpanzee DNA: A C G T G C

Only one difference.

This suggests:

- They are closely related.
- They may share a common ancestor.

Sequence Alignment (Global and Local)

Global Alignment	Local Alignment
Target Sequence 5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3' 5' ACTACTAGATT----ACGGATC--GTACTTTAGAGGCTAGCAACCA 3' Query Sequence	Target Sequence 5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3' GAGGCTTGCAACCA 3' Query Sequence
Tries to align entire sequence.	Align regions with highest similarities.
Align all letters from query and target.	Align substring (Local region) of target with substring of query.
Suitable for closely related sequences.	Suitable for divergent sequences.
A general global alignment method is Needleman-Wunsch .	A general local alignment method is Smith-Waterman .

Biological Significance of Gaps

- The biological processes involved in **protein synthesis** — namely **replication, transcription, and translation** — can sometimes produce errors.
- These errors lead to **mutations** in the resulting **nucleic acid sequence** (DNA or RNA), which may alter the structure or function of the synthesized protein.

Types of Mutations

Mutational changes in sequences are generally classified into **three main types**:

1. Substitution

- One base (nucleotide) is replaced by another.
- In sequence alignment, **scoring matrices** such as **PAM (Point Accepted Mutation)** or **BLOSUM (Blocks Substitution Matrix)** are used to **score substitutions** between amino acids or nucleotides.

2. Insertion

- One or more extra bases are **added** into the sequence.
- This may shift the reading frame if it occurs within a coding region.

3. Deletion

- One or more bases are **removed** from the sequence.
- Like insertion, it can also cause a mutation.

Sequence Alignment

- Terms Used in Sequence Alignment/ Comparison:

(Qry)	A C D E F G	A C D E F G	A C D E F G	A C -- E F G
(Sbj)	A C D E F G	A C L E F G	A C -- E F G	A C D E F G

Biological event

Conservation

Substitution

Insertion

Deletion

Alignment represent

Match

Mismatch

Gap

Gap

- **Alignment representation:** Match
- No evolutionary change occurred.
- The residue is conserved.

one residue changed during evolution.

A residue was inserted into the query sequence

A residue was deleted from the query sequence

Biological Significance of Gaps

Insertion and Deletion Handling in Alignment

In sequence alignment, insertions and deletions are handled using gap penalties.

Gap opening penalty: Applied when a new gap is introduced.

Gap extension penalty: Applied when an existing gap is extended.

These penalties ensure that gaps are introduced **only when necessary** to achieve the best possible alignment.

- **What is Gap penalty?**
It is the cost to introduce gap.

Biological Significance of Gaps

Gap Penalty

A gap penalty is the cost assigned to introducing a gap (insertion or deletion) in a sequence alignment.

Gaps are not in nature. Gap is what we do in order to understand evolutionary relationships between sequences.

In order to incorporate gaps, we have gap penalty for each gap in the sequence.

We must be very careful in adding gaps in the sequence. It would be good to have 1 mismatch than to have many gaps.

Biological Significance of Gaps

Gaps can occur at various different places in a sequence as depicted to the diagram right.

However, assigning penalty values can be more or less arbitrary because there is no evolutionary theory to determine a precise cost for introducing insertions and deletions.

If the penalty values are set too low, gaps can become too numerous to allow even unrelated sequences to be matched up with high similarity scores.

Before the first character of a string

```
CTGCGGG---GGTAAT
| | | |   | | |
--GCGG-AGAGG-AA-
```

Inside a string

```
CTGCGGG---GGTAAT
| | | |   | | |
--GCGG-AGAGG-AA-
```

After the last character of a string

```
CTGCGGG---GGTAAT
| | | |   | | |
--GCGG-AGAGG-AA-
```

Biological Significance of Gaps

- If the penalty values are set too high, gaps can become too difficult to appear, and reasonable alignment cannot be achieved, which is also unrealistic.

**When the gap penalty is high,
fewer gaps will be inserted.**

If you are searching for sequences that are a strict match for your query sequence, the gap penalty should be set high. This will retrieve regions with very closely related sequences.

**When the gap penalty is low,
more and larger gaps will be
inserted.**

If you are searching for similarity between distantly related sequences, the gap penalty should be set low.

- **High gap penalty → strict alignment → closely related sequences**
- **Low gap penalty → flexible alignment → distantly related sequences**
- **So the gap value is selected to produce the most biologically realistic alignment.**

Types of Gap Penalty

- The three main types of gap penalties are: **constant**, **linear** and **affine gap penalty**.
- **Constant Gap Penalty:**
 - This is the simplest type of gap penalty.
 - Assign the same negative score for each gap position regardless whether it is opening or extending.'
 - However, this penalty scheme has been found to be less realistic.

Aligning two short DNA sequences, with “-” depicting a gap of one base pair. If each match was worth 1 point and the gap -1, the total score: $7 - 1 = 6$

```
ATTGACCTGA
| |       | | | |
AT - - -CCTGA
```

Types of Gap Penalty

- **Linear Gap Penalty:**

- The linear gap penalty takes into account the length (L) of each insertion/deletion in the gap.
- Therefore, if the penalty for each inserted/deleted element is B and the length of the gap is L, then total gap penalty = $B \cdot L$.
- However, this penalty scheme has been found to be less realistic.

Unlike constant gap penalty, the size of the gap is considered.
With a match with score 1 and gap -1, the total score: $7 - 3 = 4$

```
ATTGACCTGA
| |       | | | |
AT - - -CCTGA
```

Types of Gap Penalty

Consider two sequences:

Sequence 1: A T G C A T G

Sequence 2: A T – – A T G

Scoring scheme:

Match = +1

Gap penalty per position (B) = -1

Step-by-step calculation:

Number of matches = 5 $\rightarrow$ score = $5 \times 1 = 5$

Gap length (L) = 2

Total gap penalty = $-1 \times 2 = -2$

Total alignment score:

$5 - 2 = 3$

Types of Gap Penalty

- **Affine Gap Penalty:**

- The most widely used gap penalty function is the affine gap penalty.
- The affine gap penalty combines the components in both the constant and linear gap penalty. For example, one may use a -12/-1 scheme in which the gap opening penalty is -12 and the gap extension penalty is -1.

Formula

$$\text{Gap penalty} = G + (L - 1) \times E$$

Where:

G = gap opening penalty

E = gap extension penalty

L = length of the gap

Types of Gap Penalty

Example of Affine Gap Penalty

Given sequences

Sequence 1: A C G T T A

Sequence 2: A C – – T A

Scoring scheme

Match = +2

Mismatch = -1

Gap opening penalty (G) = -3

Gap extension penalty (E) = -1

Step 1: Identify gaps

One gap of length $L = 2$

Types of Gap Penalty

Step 2: Calculate affine gap penalty

$$\begin{aligned}\text{Gap penalty} &= G + (L-1) \times E \\ &= (-3) + (2-1) \times (-1) = -4\end{aligned}$$

Step 3: Calculate match score

$$\begin{aligned}\text{Matches: A-A, C-C, T-T, A-A} &\rightarrow 4 \text{ matches} \\ 4 \times 2 &= 8\end{aligned}$$

Step 4: Total alignment score

$$\text{Total score} = 8 - 4 = 4$$

Affine Gap Penalty

1

-5 -5

AGGCTACT**T**~**T**~**T**CA
GGCTACT**T****A****T****A****T**CA

Gap penalty: -10

2

-5 -1

AGGCTACT**T****T****T**~~CA
GGCTACT**T****A****T****A****T**CA

Gap penalty: -6

Scoring Alignment

Why Scoring Alignment!

- We need to differentiate good alignments from poor ones.
- We use a rule that assigns a numerical score to any alignment; the higher the score, the better the alignment.
- Sequence Similarity and Identity
- Sequence Alignment Score

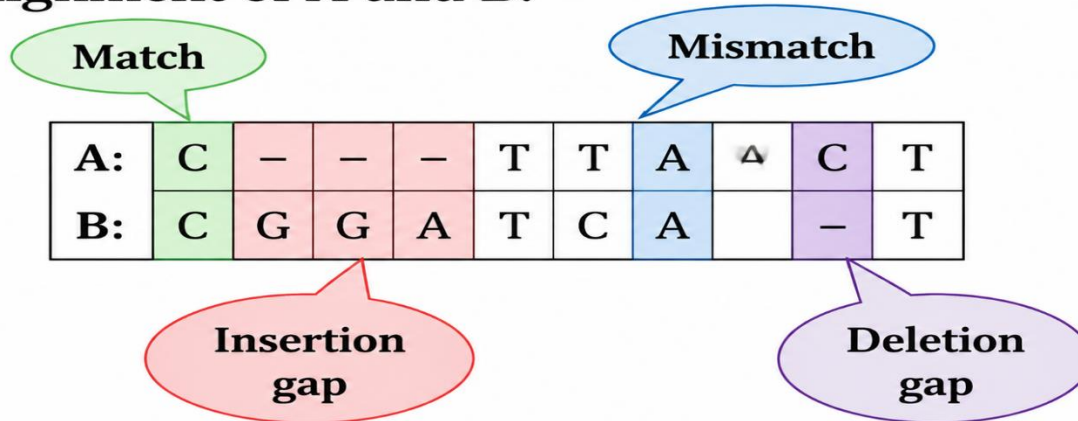
Sequence Alignment Score

- Sequence similarity score is calculated based on the pairwise alignment quality.
- Alignment score is the sum of scores for each position.

Sequence A: **CTTAACT**

Sequence B: **CGGATCAT**

An alignment of A and B:



Sequence Alignment Score (Example-1)

AACGTTTCTCCAAATAGCTAGGC

=== - - ===== - ===== - == - =====

AACCGTTCTACAATTACCTAGGC

Hits (+1) : 18

Misses (-2) : 5

Score = $18 \times 1 + 5 \times (-2) = 8$

Sequence Alignment Score (Example-2)

- Match: **+8**
- Mismatch: **-5**
- Each gap symbol: **-3**

C	-	-	-	T	T	A	A	C	T
C	G	G	A	T	C	A	-	-	T
+8	-3	-3	-3	+8	-5	+8	-3	-3	+8

= +12

Sequence Alignment Score (Example-3)

Alignment of ATCGGATCT and ACGGACT

A	T	C	G	G	A	T	C	T
A	-	C	G	G	-	A	C	T

match: +2

mismatch: -1

indel: -2

6 matches, 1 mismatch, and 2 indels

$$6 \times 2 + 1 \times (-1) + 2 \times (-2) = 7$$

Sequence Alignment Score (Example-4)

- match: **+4**

- mismatch: **-5**

- gap opening: **-16**

A	C	G	A	C	T	G	G	C	A
		×		-					
A	C	T	A		T	G	G	C	A

- **4 + 4 + (-5) + 4 + (-16) + 4 + 4 + 4 + 4 + 4 = 11**

Sequence Alignment Scoring Scheme

Substitution score matrix

	a	c	g	t
a	2	-3	-1	-3
c	-3	2	-3	-1
g	-1	-3	2	-3
t	-3	-1	-3	2

Gap scores

Gap existence cost: 5
Gap extension cost: 1

Alignment score = 11

Example:	t	a	c	g	t	g	-	-	a	g	g	t
	t	a	c	a	t	g	c	t	a	g	g	t
Score:	2	+2	+2	-1	+2	+2	-5	-1	+2	+2	+2	+2

Global Alignment

Global Alignment

Rewards and Penalties

Match : 1

Mismatch : -1

GAP : -2

Dynamic Programming

P

R

O

B

L

E

M

STEPS

Initialization



Matrix Filling



Traceback

Needleman and Wunch Algorithm Example

		A	T	G	C	T
A						
G						
C						
T						

Rewards and Penalties

Match : 1 
Mismatch : -1 
GAP : -2 

Sequences

Seq 1 = A T G C T

Seq 2 = A G C T

Needleman and Wunch Algorithm

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2					
G	-4					
C	-6					
T	-8					

Rewards and Penalties

Match : 1 
Mismatch : -1 
GAP : -2 

Sequences

Seq 1 = A T G C T

Seq 2 = A G C T

Needleman and Wunch Algorithm

Match : 1 
Mismatch : -1 
GAP : -2 

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2					
G	-4					
C	-6					
T	-8					

	A
A	0
A	-2

Value form LEFT -

Value form UP -

Value form DIAGONAL -

Needleman and Wunch Algorithm

Match : 1 
Mismatch : -1 
GAP : -2 

A = A
MATCHED

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2					
G	-4					
C	-6					
T	-8					

	A
A	0
A	-2



Value form LEFT - **-4**
Value form UP - **-4**
Value form DIAGONAL - **1**

Maximum = **1**

Needleman and Wunch Algorithm

Match : 1 ■
 Mismatch : -1 ■
 GAP : -2 ■

A ≠ T
MISMATCHED

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1			
G	-4					
C	-6					
T	-8					

T

-2	-4
1	

A

DIAGONAL

Value form LEFT - **-1**

Value form UP - **-6**

Value form DIAGONAL - **-3**

Maximum = **-1**

Needleman and Wunch Algorithm

Match : 1 
Mismatch : -1 
GAP : -2 

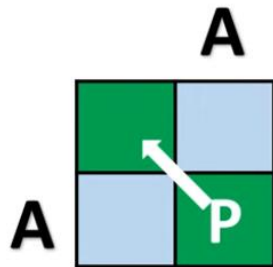
		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2

Traceback

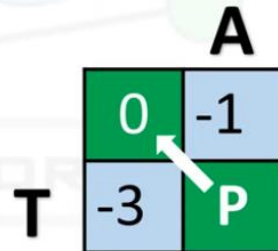
Beginning

From the right - bottom in the matrix where the highest score is present, continuing upto the upper left corner

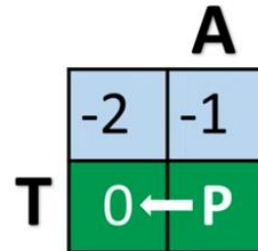
You can trace the arrows back that lead to the starting point



MATCHED



NOT MATCHED



Trace back

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2

Starting from highest score
i.e in the right most bottom

T = T

Trace back

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2

C = C

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2



$G = G$

ATICA

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2

$A \neq T$

	-1

1

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2

A = A

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
G	-4	-1	0	0	-2	-4
C	-6	-3	-2	-1	1	-1
T	-8	-5	-2	-3	-1	2

Seq-1

A T G C T

Seq-2

A - G C T



BIOINFORMATICA

HAVE A TRY

Seq 1 = AATCG
Seq 2 = AACG

Seq 1 = CCGTCG
Seq 2 = CCGCG

Seq 1 = CGCTCGCT
Seq 2 = CACTCGT

Seq1: CTAGC
Seq2: CAGTC

Match Score: +2
Mismatch Score: -1
Gap Penalty: -3

	C	T	A	G	C
C					
A					
G					
T					
C					

Final Alignment

NEEDLEMAN WUNSCH ALGORITHM

		T	A	T	G	A
T A C G A	0	-2	-4	-6	-8	-10
	-2					
	-4					
	-6					
	-8					
	-10					

MATCH = 1
MISMATCH = -1
GAP = -2

	T	A	T	G	A	
T	0	-2	-4	-6	-8	-10
A	-2	1	-1	-3	-5	-7
C	-4	-1	2	0	-2	-4
G	-6	-3	0	1	-1	-3
A	-8	-5	-2	-1	2	0
A	-10	-7	-4	-3	0	3

Global pairwise alignment using Needleman-Wunsch

(f)

Sequence 2

Sequence 1

		F	M	D	T	P	L	N	E
	0	-2	-4	-6	-8	-10	-12	-14	-16
F	-2	1	-1	-3	-5	-7	-9	-11	-13
K	-4	-1							
H	-6								
M	-8								
E	-10								
D	-12								
P	-14								
L	-16								
E	-18								

(g)

Sequence 2

Sequence 1

		F	M	D	T	P	L	N	E
	0	-2	-4	-6	-8	-10	-12	-14	-16
F	-2	1	-1	-3	-5	-7	-9	-11	-13
K	-4	-1	-1	-3	-5	-7	-9	-11	-13
H	-6	-3	-3	-3	-5	-7	-9	-11	-13
M	-8	-5	-2	-4	-5	-7	-9	-11	-13
E	-10	-7	-4	-4	-6	-7	-9	-11	-10
D	-12	-9	-6	-3	-5	-7	-9	-11	-12
P	-14	-11	-8	-5	-5	-4	-6	-8	-10
L	-16	-13	-10	-7	-7	-6	-3	-5	-7
E	-18	-15	-12	-9	-9	-8	-5	-5	-4

Global pairwise alignment using Needleman-Wunsch

(a)

Sequence 2

Sequence 1

		F	M	D	T	P	L	N	E
	0	-2	-4	-6	-8	-10	-12	-14	-16
F	-2	1	-1	-3	-5	-7	-9	-11	-13
K	-4	-1	-1	-3	-5	-7	-9	-11	-13
H	-6	-3	-3	-3	-5	-7	-9	-11	-13
M	-8	-5	-2	-4	-5	-7	-9	-11	-13
E	-10	-7	-4	-4	-6	-7	-9	-11	-10
D	-12	-9	-6	-3	-5	-7	-9	-11	-12
P	-14	-11	-8	-5	-5	-4	-6	-8	-10
L	-16	-13	-10	-7	-7	-6	-3	-5	-7
E	-18	-15	-12	-9	-9	-8	-5	-5	-4

Highlighted cells indicate the optimal path (best scores), indicating how the two sequences should be aligned.

(b)

Sequence 2

		F	M	D	T	P	L	N	E	
Sequence 1		0	-2	-4	-6	-8	-10	-12	-14	-16
	F	-2	1	-1	-3	-5	-7	-9	-11	-13
	K	-4	-1	-1	-3	-5	-7	-9	-11	-13
	H	-6	-3	-3	-3	-5	-7	-9	-11	-13
	M	-8	-5	-2	-4	-5	-7	-9	-11	-13
	E	-10	-7	-4	-4	-6	-7	-9	-11	-10
	D	-12	-9	-6	-3	-5	-7	-9	-11	-12
	P	-14	-11	-8	-5	-5	-4	-6	-8	-10
	L	-16	-13	-10	-7	-7	-6	-3	-5	-7
	E	-18	-15	-12	-9	-9	-8	-5	-5	-4

(b)

Sequence 1

		Sequence 2								
		F	M	D	T	P	L	N	E	
Sequence 1		0	-2	-4	-6	-8	-10	-12	-14	-16
	F	-2	+1	-1	-3	-5	-7	-9	-11	-13
	K	-4	-1	-1	-3	-5	-7	-9	-11	-13
	H	-6	-3	-3	-5	-7	-9	-11	-13	-13
	M	-6	-5	-2	-4	-5	-7	-9	-11	-13
	E	-10	-7	-4	-4	-6	-7	-9	-11	-10
	D	-12	-9	-6	-3	-5	-7	-9	-11	-12
	P	-14	-11	-8	-5	-5	-4	-6	-8	-10
	L	-16	-13	-10	-7	-7	-6	-3	-5	-7
	E	-18	-15	-12	-9	-9	-8	-5	-5	-4

✓ Sequence 1

✓ Sequence 2



F	K	H	M	E	D	-	P	L	-	E
F	-	-	M	-	D	T	P	L	N	E

LOCAL ALIGNMENT

Target Sequence

5' ACTACTAGATTACTTACGGATCAGGTACTTTAGAGGCTTGCAACCA 3'

||||| ||||| ||||| ||||| |||||

Query Sequence

5' TACTCACGGATGAGGTACTTTAGAGGC 3'

Align regions having highest similarities between 2 sequences

Stretches of sequences with highest density of matches are aligned

More suitable for partially similar , different length and conserved regions containing sequences

Align substring of target sequence with substrings of query sequence

Suitable for divergent sequences

Most general local alignment algorithm is the

Smith Waterman

SMITH WATERMAN

NEGATIVE VALUES

BECOMES

0

Rewards and Penalties

Mismatch

-1

Match

1

Gap

-2

		A	T	G	C	T
	0	-2	-4	-6	-8	-10
A	-2					
G	-4					
C	-6					
T	-8					

Negative → 0

		A	T	G	C	T
	0	0	0	0	0	0
A	0					
G	0					
C	0					
T	0					

		A	T	G	C	T
	0	0	0	0	0	0
A	0					
G	0					
C	0					
T	0					

Negative



0

-2

-2

1

		A	T	G	C	T
	0	0	0	0	0	0
A	0					
G	0					
C	0					
T	0					

0

0

1

		A	T	G	C	T
	0	0	0	0	0	0
A	0	1				
G	0					
C	0					
T	0					

		A	T	G	C	T
	0	0	0	0	0	0
A	0	1	0			
G	0					
C	0					
T	0					

Negative → 0

-1

-2

-1

TRACE BACK

Start from
Highest Value

Proceed towards
Zeros

An arrow
for
a base

Multiple local
alignments
possible

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0
0	0	2	0	1	0	0	0
0	0	0	0	0	2	0	0
0	0	0	0	0	0	3	0
0	0	0	0	0	0	0	4
0	0	0	0	0	0	0	0

4 3 2 1 0

&

2 1 0

		A	T	G	C	T
	0	0	0	0	0	0
A	0	1	0	0	0	0
G	0	0	0	1	0	0
C	0	0	0	0	2	0
T	0	0	0	0	0	3

		A	T	G	C	T
	0	0	0	0	0	0
A	0	1	0	0	0	0
G	0	0	0	1	0	0
C	0	0	0	0	2	0
T	0	0	0	0	0	3

GCT



GCT

Locally Aligned Sequences

Sequence 1

ATGCT

Sequence 2

AGCT

| | |

Seq 1

	-	G	A	A	T	T	C
-	0	0	0	0	0	0	0
G	0	2	0	0	0	0	0
A	0	0	4	2	0	0	0
T	0	0	2	3	4	2	0
A	0	0	0	4	2	3	1
C	0	0	0	2	3	1	5

Seq 2

Seq 1: G A A T T C

Seq 2: G A - T A C

$$2 + 2 - 2 + 2 - 1 + 2 = 5$$

An other Example:

Perform Local alignment using Dynamic programming:

Eg. Given Seq 1: GAATTCAT and Seq 2: CCTCATG and

Scores: starting score=0, match=2, mismatch = -1 and gap= -2.

Seq 1

	-	G	A	A	T	T	C	A	T
0	0	0	0	0	0	0	0	0	0
C	0	0	0	0	0	0	2	0	0
C	0	0	0	0	0	0	2	1	0
T	0	0	0	0	2	2	0	1	3
C	0	0	0	0	0	1	4	2	1
A	0	0	2	2	0	0	2	6	4
T	0	0	0	1	4	2	0	2	8
G	0	2	0	0	2	3	1	0	6

Seq 1: T C A T
 | | | |
Seq 2: T C A T

$$2+2+2+2 = 8$$